

# Clinical Risk Assessment Report: Patient 19

## 1. Primary Outcome & Risk Summary

- Target: Occult Lymph Node (OLN) Metastasis
- Model Prediction: Negative for Metastasis
- Prediction Confidence: Moderate Confidence
- Absolute Predicted Risk: 11.6% (Diagnostic Threshold: 22.1%)
- Relative Risk Multiplier: 0.53x the diagnostic threshold

### Clinical Takeaway:

The machine learning model classified this patient as Low-Risk for Occult Lymph Node (OLN) Metastasis. The primary driver determining this prediction is the Large Zone Emphasis (CT) (GLZLM\_LGZE.CT), which reduced the patient's absolute risk by 1.8%. Biologically, this feature reflects the presence of large homogeneous areas; lower values suggest a chaotic, heterogeneous microenvironment.

## 2. Key Pathological & Imaging Drivers (Elevating Risk)

**Squamous Cell Carcinoma Antigen (SCC\_Ag)** Value: 0.7 [Avg (28th %ile)] (+1.5%)  
*Serum levels reflect tumor proliferation, burden, and squamous cell differentiation.*

## 3. Protective / Mitigating Factors (Reducing Risk)

**Large Zone Emphasis (CT) (GLZLM\_LGZE.CT)** Value: -0.18 [Avg (54th %ile)] (-1.8%)  
*Reflects the presence of large homogeneous areas; lower values suggest a chaotic, heterogeneous microenvironment.*

**Large Zone Low Gray-Level Emphasis (CT) (GLZLM\_LZLGE.CT)** Value: -0.4 [Bottom <1%] (-1.1%)  
*Indicates the dominance of large regions with low attenuation or metabolism, pointing to extensive necrosis or ground-glass opacities.*

**Long Run High Gray-Level Emphasis (PET) (GLRLM\_LRHGE.PET)** Value: -0.56 [Avg (28th %ile)] (-1.1%)  
*Points to continuous, elongated bands of high-density or highly active tumor tissue.*

**Short Run Low Gray-Level Emphasis (CT) (GLRLM\_SRLGE.CT)** Value: -0.13 [Avg (54th %ile)] (-0.8%)  
*Indicates fragmented areas of low density, often associated with diffuse micro-necrosis or loose tissue structure.*

**Minimum Metabolic Activity (SUVmin) (CONVENTIONAL\_SUVbwmin)** Value: -0.85 [Bottom 17%] (-0.7%)  
*Indicates the areas of lowest metabolic activity within the tumor, which may represent necrotic or poorly perfused regions.*

**Maximum Tumor Density (CONVENTIONAL\_HUmax)** Value: 0.09 [Avg (48th %ile)] (-0.7%)  
*Reflects the most dense solid components or calcifications within the primary tumor lesion.*

**75th Percentile Tumor Density (CONVENTIONAL\_HUQ3)** Value: 0.27 [Avg (49th %ile)] (-0.6%)  
*Reflects the overall physical density, solid components, and general attenuation characteristics of the tumor.*

Machine Learning Feature Attribution (SHAP Decision Path)

